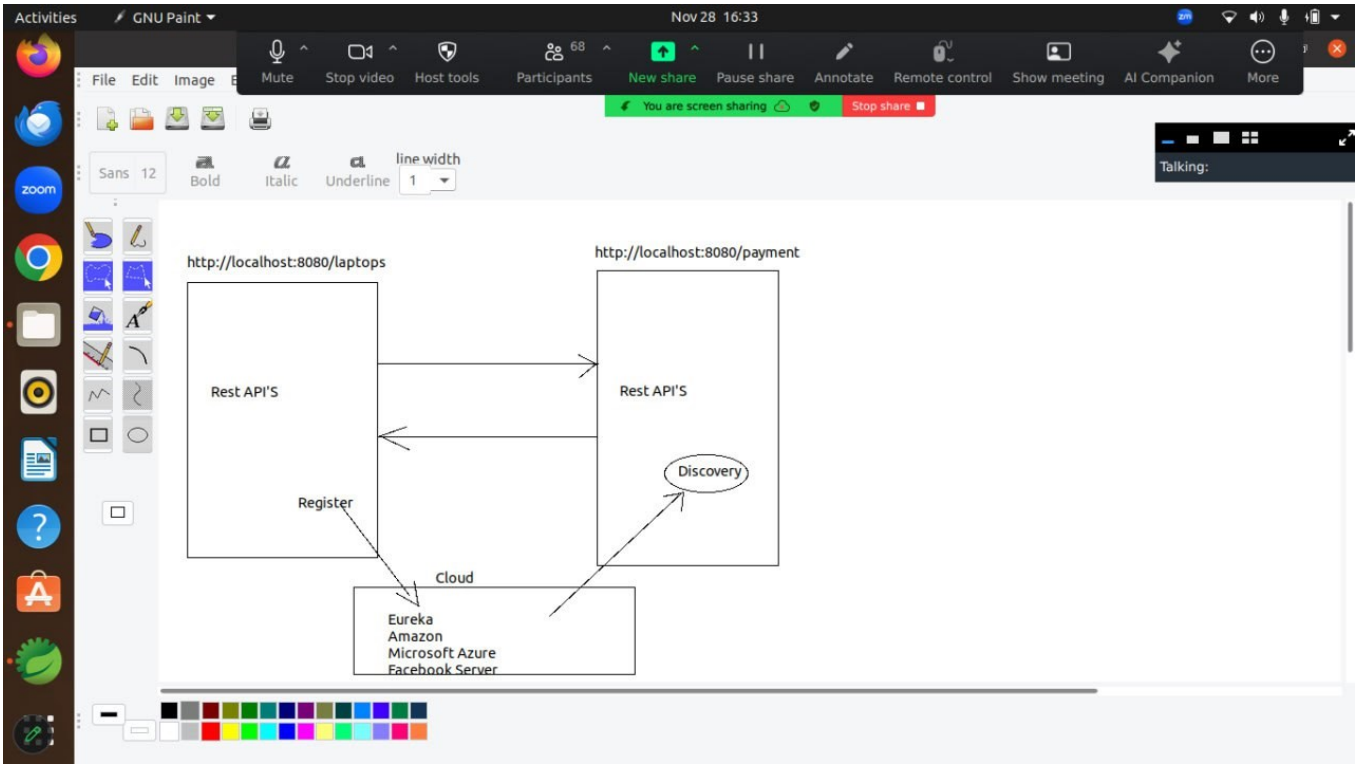
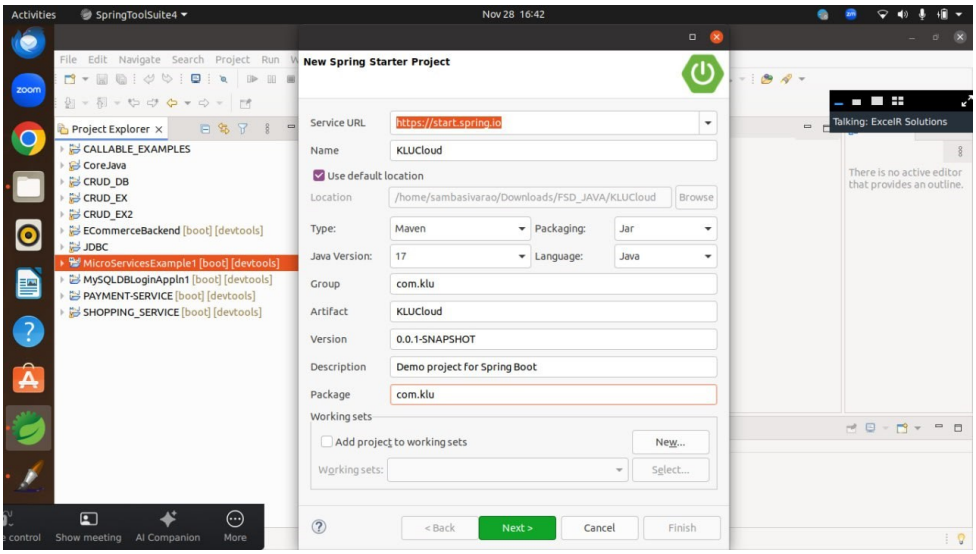
# Microservices using spring boot

## Creation of Eureka Server

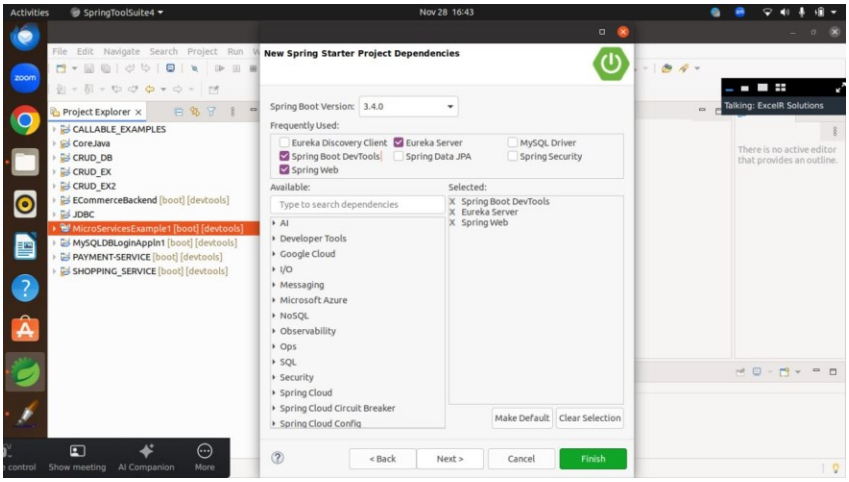
## Step-1



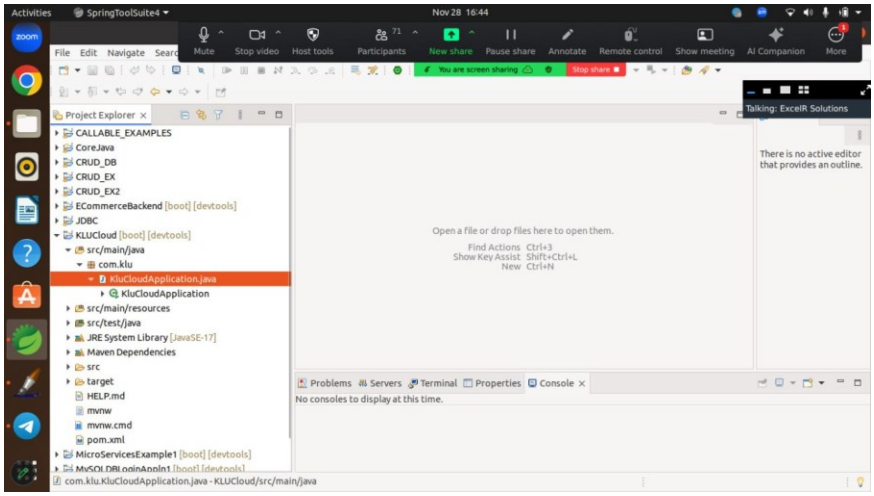

## Step-2


Activities
SpringToolSuite4
Nov 28 16:42
File Edit Navigate Search Project Run
Project Explorer
CALLABLE_EXAMPLES
CoreJava
CRUD_DB
CRUD_EX
CRUD_EX2
ECommerceBackend [boot] [devtools]
JDBC
MicroServicesExample1 [boot] [devtools]
MySQLDBLoginAppIn1 [boot] [devtools]
PAYMENT-SERVICE [boot] [devtools]
SHOPPING_SERVICE [boot] [devtools]
New Spring Starter Project
Service URL
https://start.spring.io
Name
KLUCloud
Use default location
Location
/home/sambasivarao/Downloads/FSD_JAVA/KLUCloud
Browse
Type:
Maven
Packaging:
Jar
Java Version:
17
Language:
Java
Group
com.klu
Artifact
KLUCloud
Version
0.0.1-SNAPSHOT
Description
Demo project for Spring Boot
Package
com.klu
Working sets
Add project to working sets
New...
Working sets:
Select...
< Back
Next >
Cancel
Finish
Talking: ExcelR Solutions
There is no active editor that provides an outline.
Show meeting
AI Companion
More

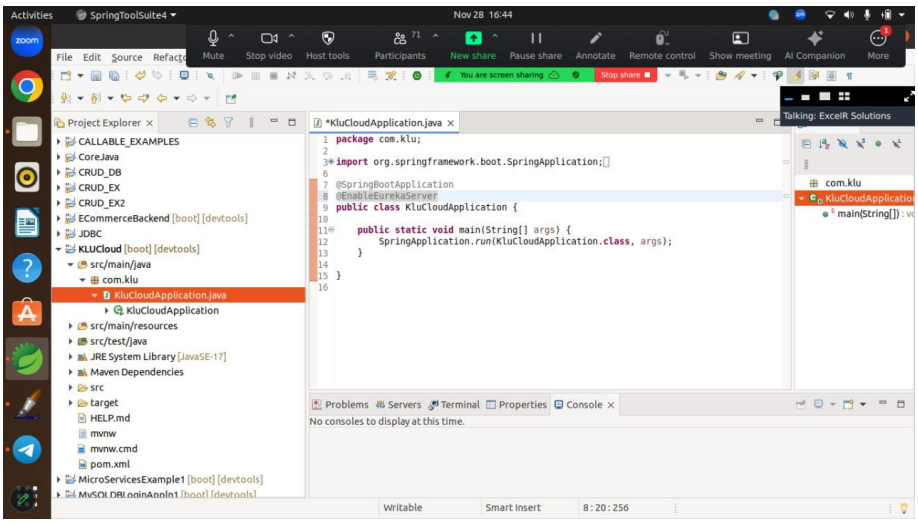
Step-3


Activities
SpringToolSuite4
Nov 28 16:43
File Edit Navigate Search Project Run
Project Explorer
CALLABLE_EXAMPLES
CoreJava
CRUD_DB
CRUD_EX
CRUD_EX2
ECommerceBackend [boot] [devtools]
JDBC
MicroServicesExample1 [boot] [devtools]
MySQLDBLoginAppIn1 [boot] [devtools]
PAYMENT-SERVICE [boot] [devtools]
SHOPPING_SERVICE [boot] [devtools]
New Spring Starter Project Dependencies
Spring Boot Version:
3.4.0
Frequently Used:
Eureka Discovery Client
Eureka Server
MySQL Driver
Spring Boot DevTools
Spring Data JPA
Spring Security
Spring Web
Available:
Type to search dependencies
AI
Developer Tools
Google Cloud
I/O
Messaging
Microsoft Azure
NoSQL
Observability
Ops
SQL
Security
Spring Cloud
Spring Cloud Circuit Breaker
Spring Cloud Config
Selected:
X Spring Boot DevTools
X Eureka Server
X Spring Web
Make Default
Clear Selection
< Back
Next >
Cancel
Finish
Talking: ExcelR Solutions
There is no active editor that provides an outline.
Show meeting
AI Companion
More

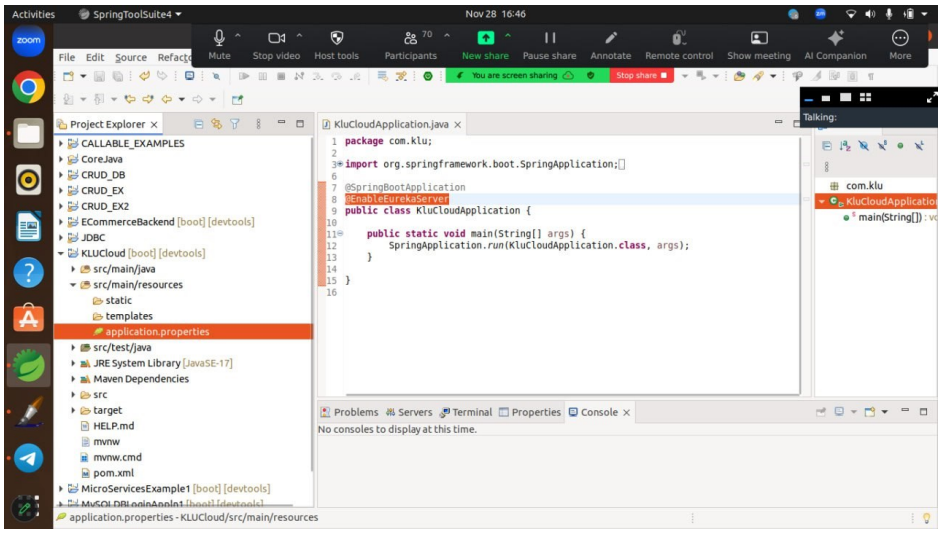
Step-4


Activities
SpringToolSuite4
Nov 28 16:44
File Edit Navigate
Mute
Stop video
Host tools
Participants
New share
Pause share
Annotate
Remote control
Show meeting
AI Companion
More
You are screen sharing
Stop share
Talking: ExcelR Solutions
Project Explorer
CALLABLE_EXAMPLES
CoreJava
CRUD_DB
CRUD_EX
CRUD_EX2
ECommerceBackend [boot] [devtools]
JDBC
KLUCloud [boot] [devtools]
src/main/java
com.klu
KluCloudApplication.java
KluCloudApplication
src/main/resources
src/test/java
JRE System Library [JavaSE-17]
Maven Dependencies
src
target
HELP.md
mvnw
mvnw.cmd
pom.xml
MicroServicesExample1 [boot] [devtools]
There is no active editor that provides an outline.
Open a file or drop files here to open them.
Find Actions Ctrl+3
Show Key Assist Shift+Ctrl+L
New Ctrl+N
Problems
Servers
Terminal
Properties
Console
No consoles to display at this time.
com.klu.KluCloudApplication.java - KLUCloud/src/main/java

Step-5


Activities
SpringToolSuite4
Nov 28 16:44
File Edit Source
Mute
Stop video
Host tools
Participants
New share
Pause share
Annotate
Remote control
Show meeting
AI Companion
More
You are screen sharing
Stop share
Talking: ExcelR Solutions
Project Explorer
CALLABLE_EXAMPLES
CoreJava
CRUD_DB
CRUD_EX
CRUD_EX2
ECommerceBackend [boot] [devtools]
JDBC
KLUCloud [boot] [devtools]
src/main/java
com.klu
KluCloudApplication.java
KluCloudApplication
src/main/resources
src/test/java
JRE System Library [JavaSE-17]
Maven Dependencies
src
target
HELP.md
mvnw
mvnw.cmd
pom.xml
MicroServicesExample1 [boot] [devtools]
*KluCloudApplication.java
package com.klu;
import org.springframework.boot.SpringApplication;
@SpringBootApplication
@EnableEurekaServer
public class KluCloudApplication {
public static void main(String[] args) {
SpringApplication.run(KluCloudApplication.class, args);
}
}
com.klu
KluCloudApplicatio
main(String[])
Problems
Servers
Terminal
Properties
Console
No consoles to display at this time.
Writable
Smart Insert
8 : 20 : 256

Step-6



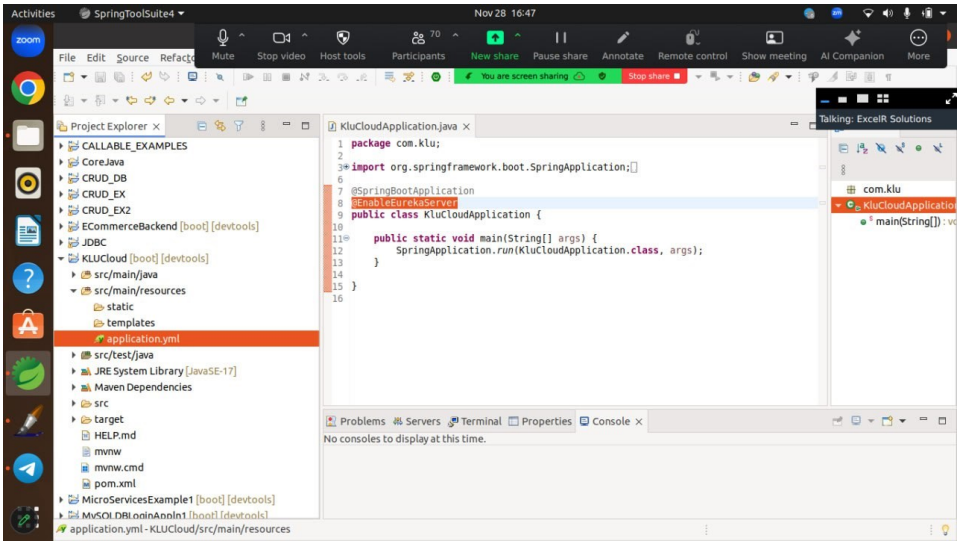

# Step-7

**Change application.properties file to application.yml**

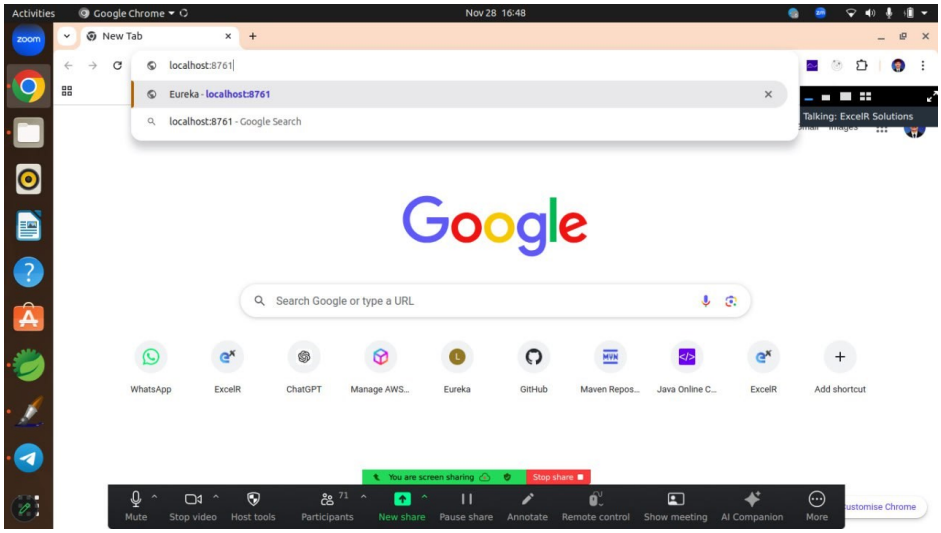
**Application.yml**

```
eureka:
  client:
    register-with-eureka: false
    fetch-registry: false

server:
  port: 8761

spring:
  application:
    name: EUREKA-CLOUD-SERVER
```

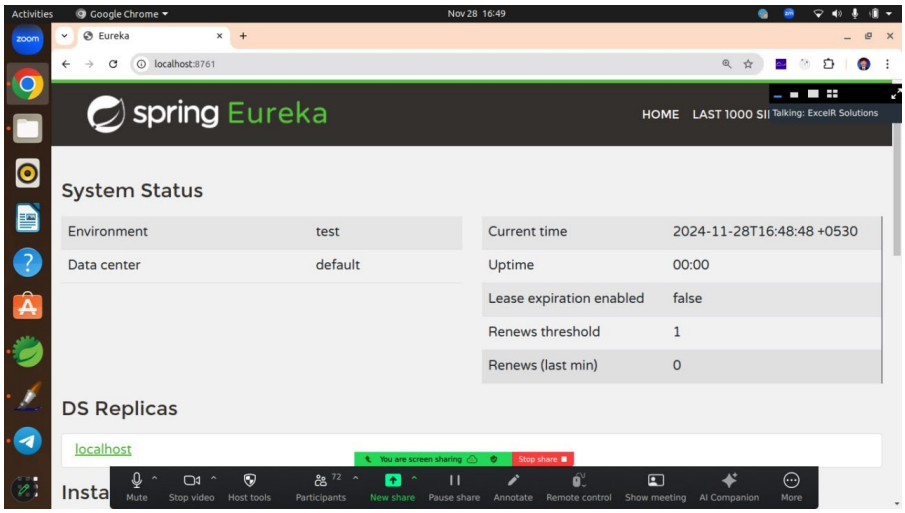
# Step-8




Step-9

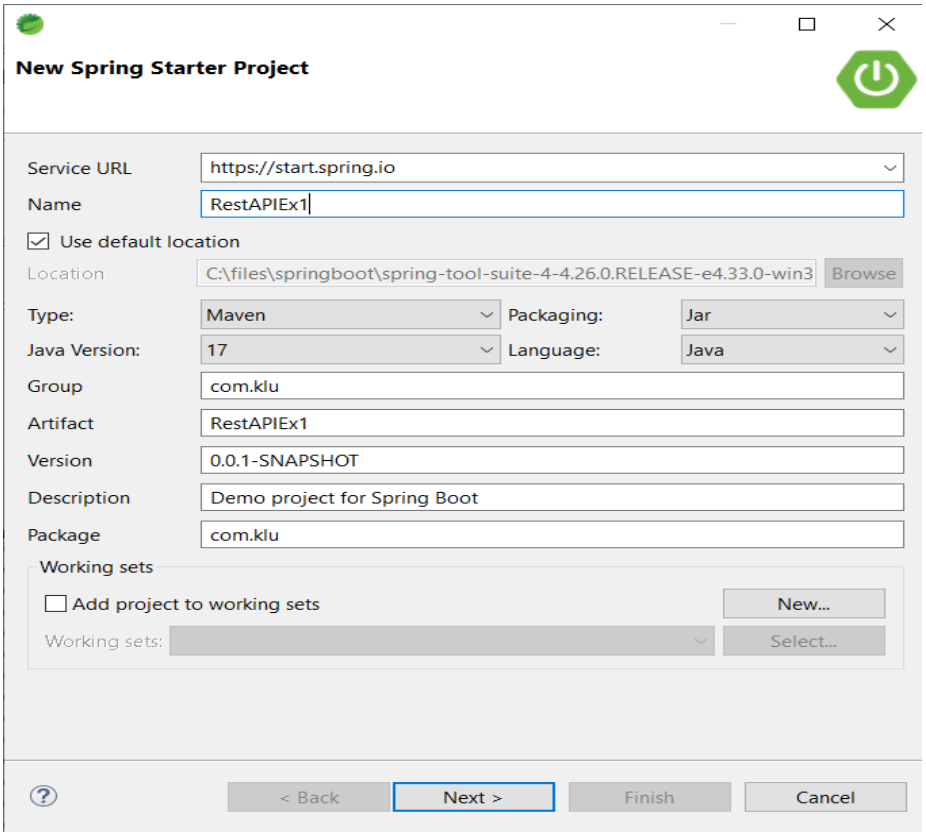



Step-10

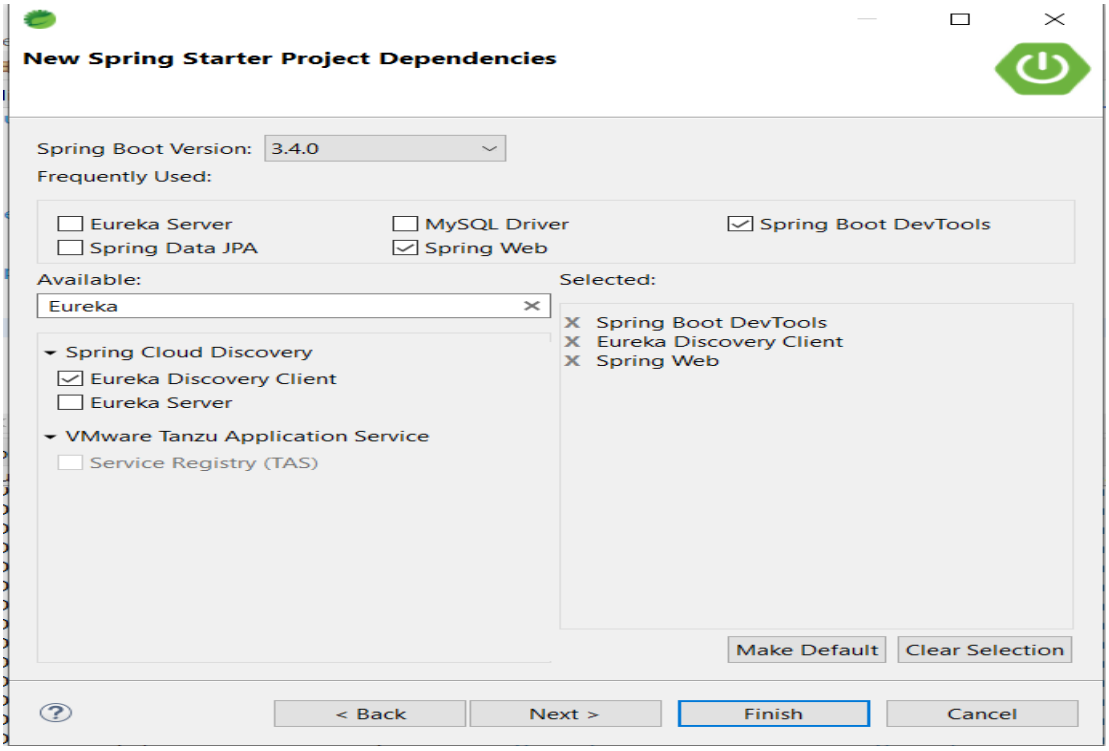

Activities
Google Chrome
Nov 28 16:49
Eureka
localhost:8761
spring Eureka
HOME
LAST 1000 SI
Talking: ExcelR Solutions
System Status
Environment
test
Data center
default
Current time
2024-11-28T16:48:48 +0530
Uptime
00:00
Lease expiration enabled
false
Renews threshold
1
Renews (last min)
0
DS Replicas
localhost
You are screen sharing
Stop share
Mute
Stop video
Host tools
Participants
New share
Pause share
Annotate
Remote control
Show meeting
AI Companion
More

# Creation RESTApi project and run in Eureka Server

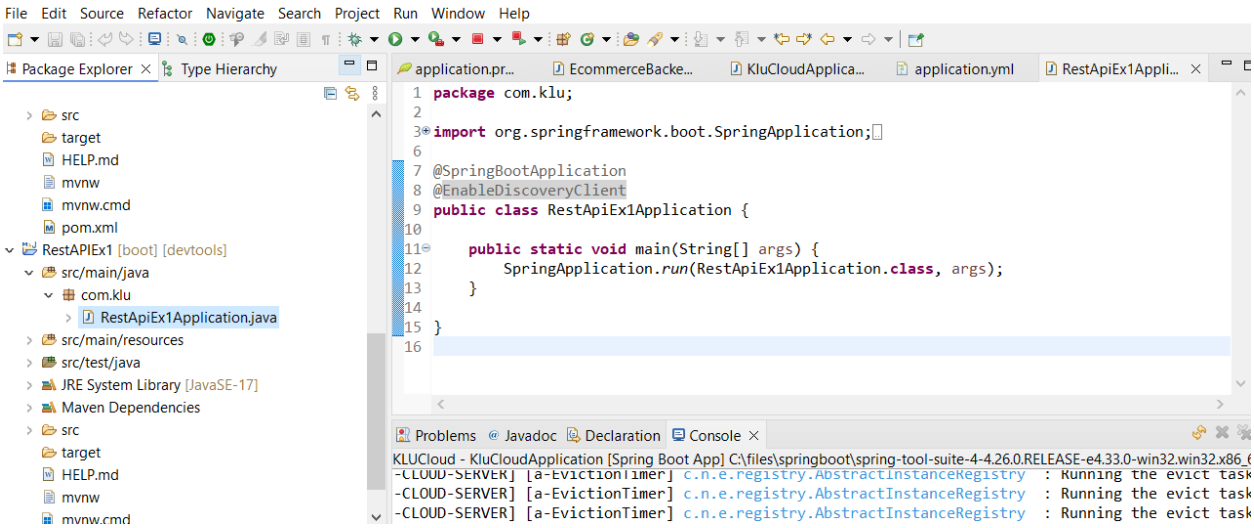
## Step-1



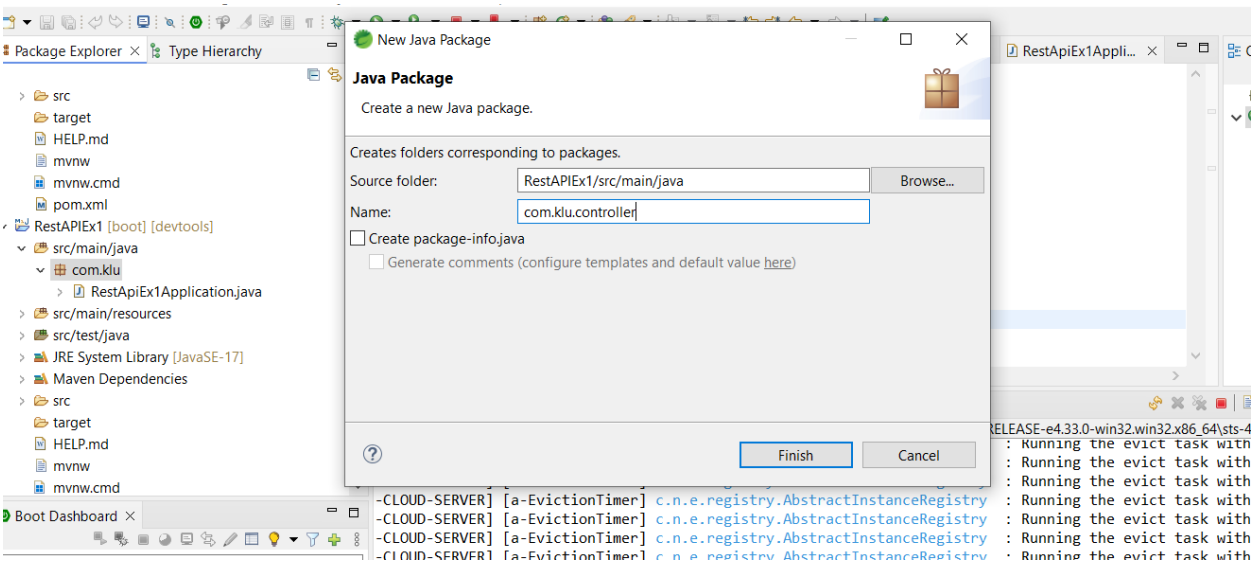

## Step-2


New Spring Starter Project Dependencies
Spring Boot Version: 3.4.0
Frequently Used:
Eureka Server
Spring Data JPA
MySQL Driver
Spring Web
Spring Boot DevTools
Available:
Eureka
Spring Cloud Discovery
Eureka Discovery Client
Eureka Server
VMware Tanzu Application Service
Service Registry (TAS)
Selected:
Spring Boot DevTools
Eureka Discovery Client
Spring Web
Make Default
Clear Selection
< Back
Next >
Finish
Cancel

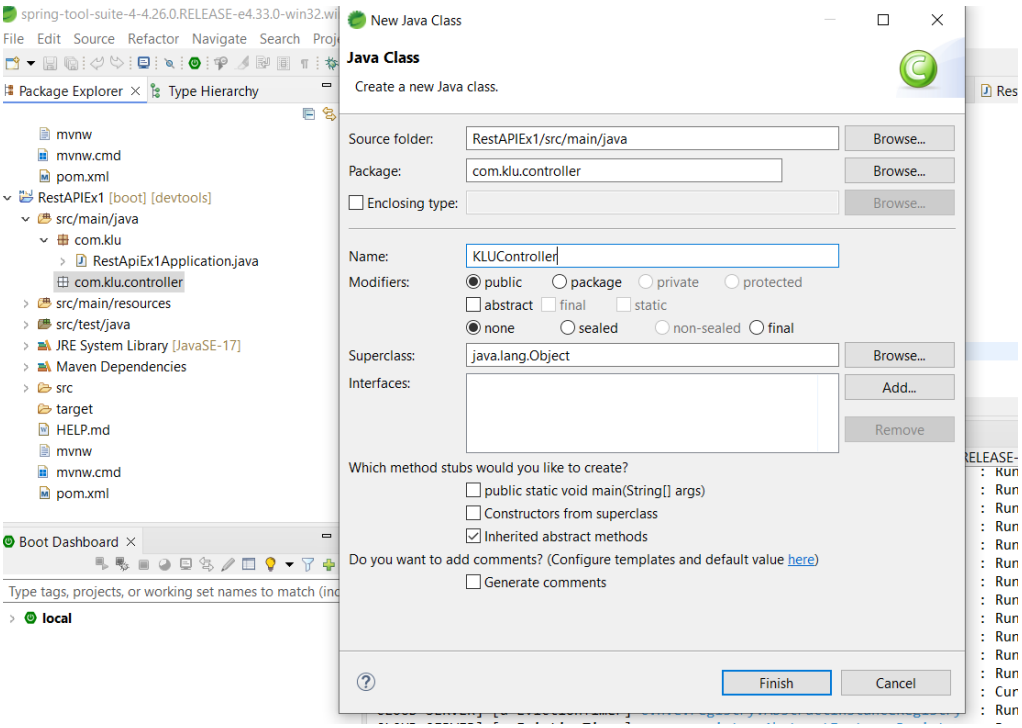
# Step-3


File Edit Source Refactor Navigate Search Project Run Window Help
Package Explorer
Type Hierarchy
src
target
HELP.md
mvnw
mvnw.cmd
pom.xml
RestAPIEx1 [boot] [devtools]
src/main/java
com.klu
RestApiEx1Application.java
src/main/resources
src/test/java
JRE System Library [JavaSE-17]
Maven Dependencies
src
target
HELP.md
mvnw
mvnw.cmd
application.pr...
EcommerceBacke...
KluCloudApplica...
application.yml
RestApiEx1Appli...
package com.klu;
import org.springframework.boot.SpringApplication;
@SpringBootApplication
@EnableDiscoveryClient
public class RestApiEx1Application {
public static void main(String[] args) {
SpringApplication.run(RestApiEx1Application.class, args);
}
}
Problems
Javadoc
Declaration
Console
KLUCloud - KluCloudApplication [Spring Boot App] C:\files\springboot\spring-tool-suite-4-4.26.0.RELEASE-e4.33.0-win32.win32.x86_6
-CLOUD-SERVER] [a-EvictionTimer] c.n.e.registry.AbstractInstanceRegistry : Running the evict task
-CLOUD-SERVER] [a-EvictionTimer] c.n.e.registry.AbstractInstanceRegistry : Running the evict task
-CLOUD-SERVER] [a-EvictionTimer] c.n.e.registry.AbstractInstanceRegistry : Running the evict task

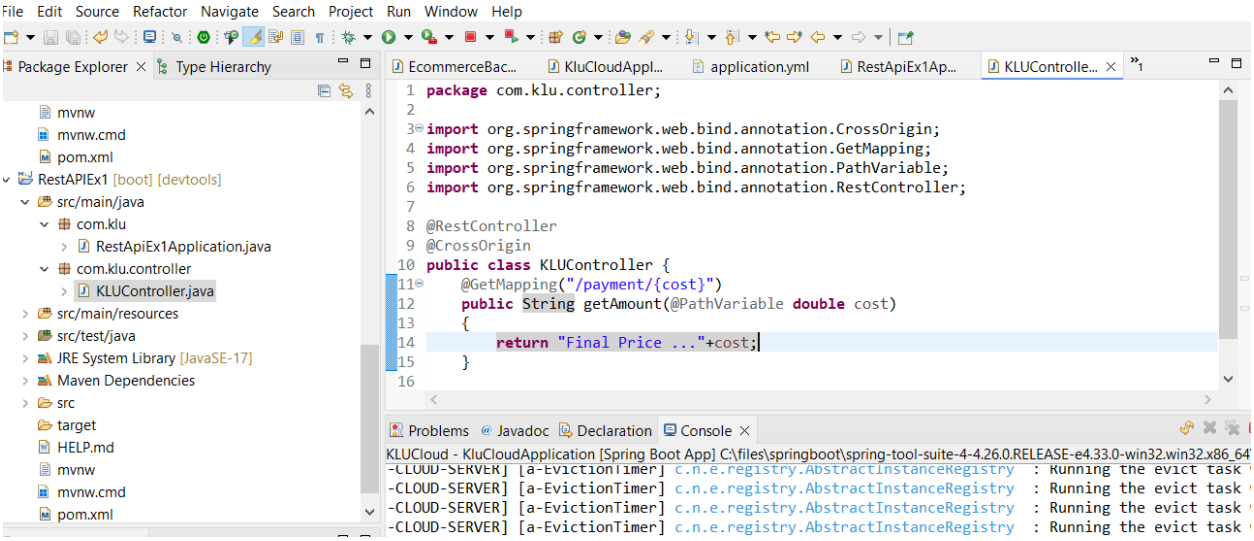
# Step-4


Package Explorer
Type Hierarchy
src
target
HELP.md
mvnw
mvnw.cmd
pom.xml
RestAPIEx1 [boot] [devtools]
src/main/java
com.klu
RestApiEx1Application.java
src/main/resources
src/test/java
JRE System Library [JavaSE-17]
Maven Dependencies
src
target
HELP.md
mvnw
mvnw.cmd
Boot Dashboard
New Java Package
Java Package
Create a new Java package.
Creates folders corresponding to packages.
Source folder:
RestAPIEx1/src/main/java
Browse...
Name:
com.klu.controller
Create package-info.java
Generate comments (configure templates and default value here)
Finish
Cancel
RestApiEx1Appli...
-CLOUD-SERVER] [a-EvictionTimer] c.n.e.registry.AbstractInstanceRegistry : Running the evict task with

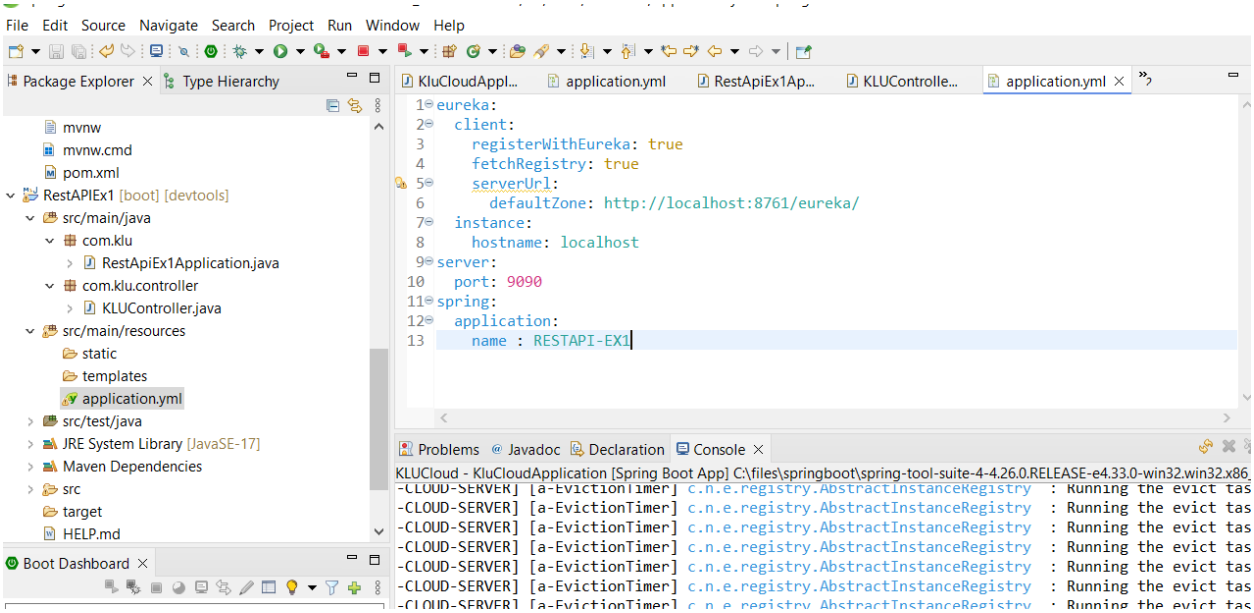
# Step-5


New Java Class
Java Class
Create a new Java class.
Source folder: RestAPIEx1/src/main/java
Package: com.klu.controller
Enclosing type:
Name: KLUController
Modifiers: public package private protected
abstract final static
none sealed non-sealed final
Superclass: java.lang.Object
Interfaces:
Which method stubs would you like to create?
public static void main(String[] args)
Constructors from superclass
Inherited abstract methods
Do you want to add comments? (Configure templates and default value here)
Generate comments
Finish
Cancel

# Step-6


package com.klu.controller;
import org.springframework.web.bind.annotation.CrossOrigin;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PathVariable;
import org.springframework.web.bind.annotation.RestController;
@RestController
@CrossOrigin
public class KLUController {
@GetMapping("/payment/{cost}")
public String getAmount(@PathVariable double cost)
{
return "Final Price ..."+cost;
}

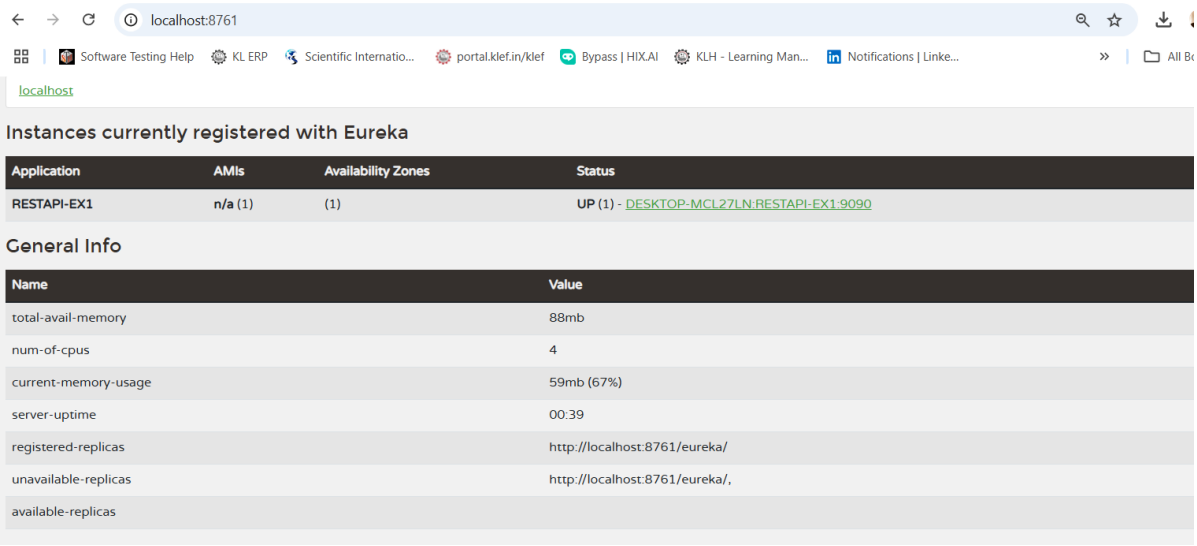
# Step-7




**Application.yml**

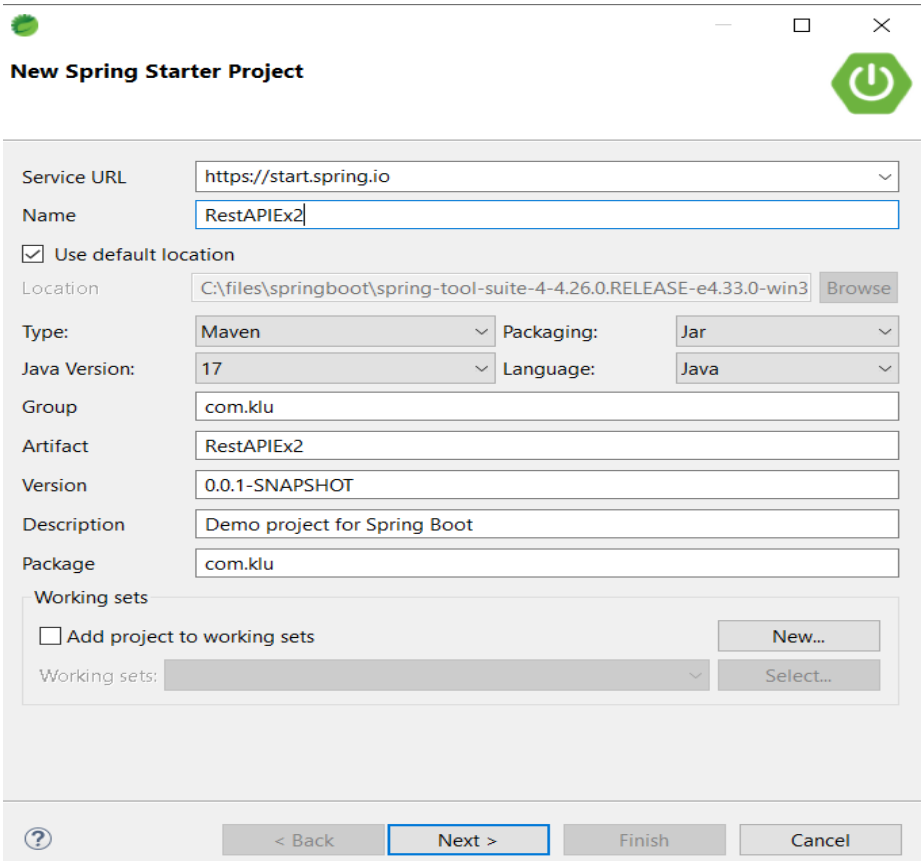
```
eureka:
  client:
    registerWithEureka: true
    fetchRegistry: true
    serverUrl:
      defaultZone: http://localhost:8761/eureka/
  instance:
    hostname: localhost
server:
  port: 8001
spring:
  application:
    name : RESTAPI-EX2
```

# Step-8

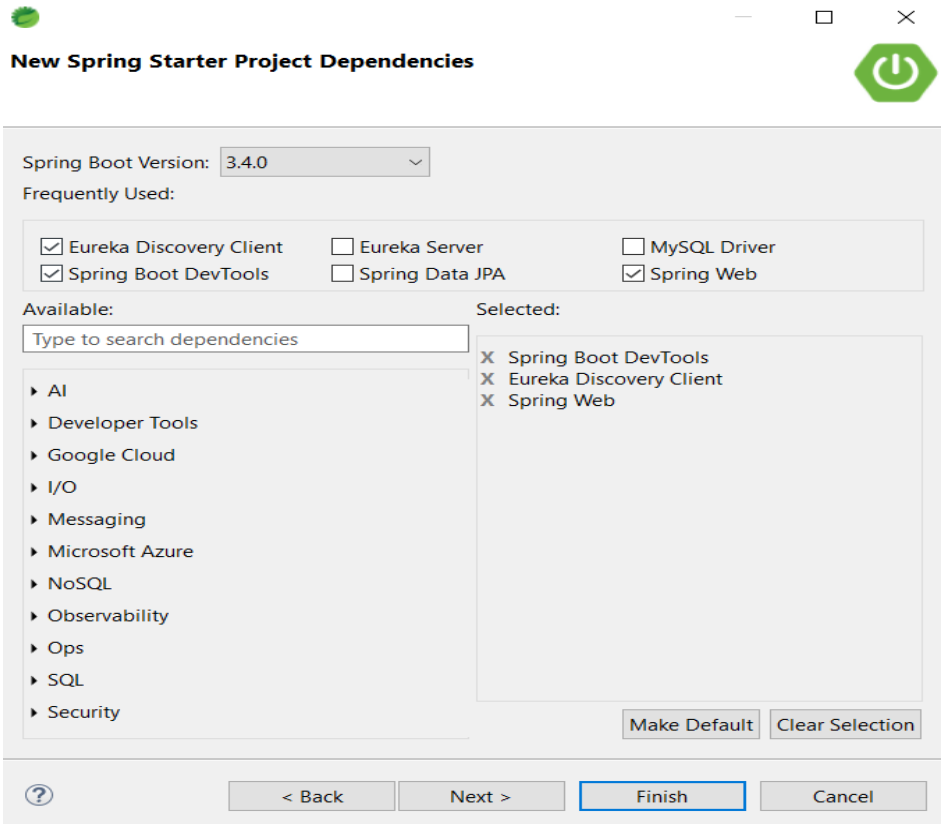

localhost:8761
Software Testing Help
KL ERP
Scientific Internatio...
portal.klef.in/klef
Bypass | HIX.AI
KLH - Learning Man...
Notifications | Linke...
localhost
Instances currently registered with Eureka
Application
AMIs
Availability Zones
Status
RESTAPI-EX1
n/a (1)
(1)
UP (1) - DESKTOP-MCL27LN:RESTAPI-EX1:9090
General Info
Name
Value
total-avail-memory
88mb
num-of-cpus
4
current-memory-usage
59mb (67%)
server-uptime
00:39
registered-replicas
http://localhost:8761/eureka/
unavailable-replicas
http://localhost:8761/eureka/,
available-replicas

# Microservices project: Deploy RestAPIEx2 Project in RestAPIEx1

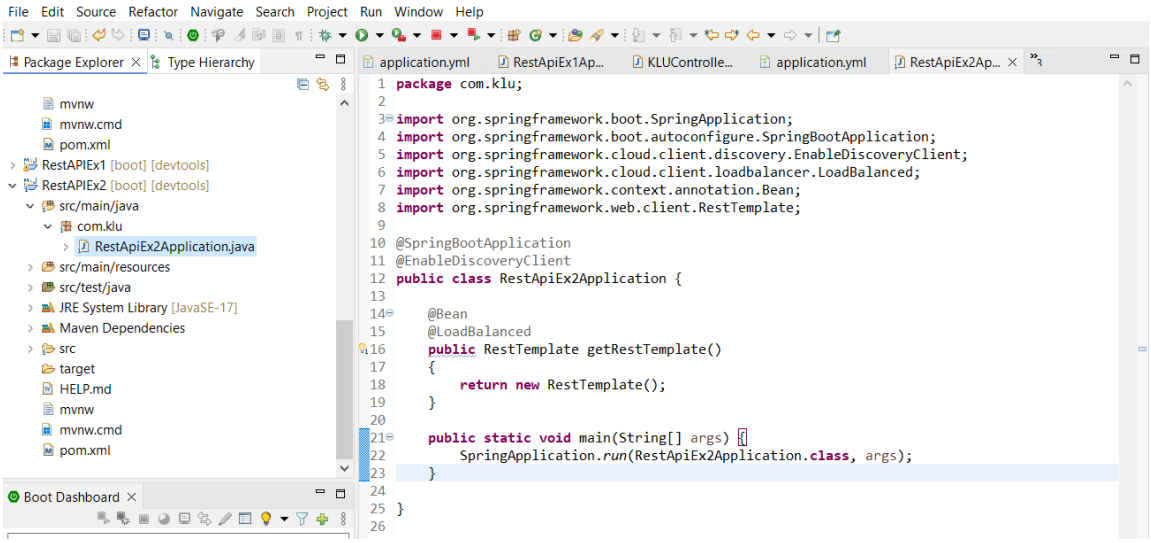
## RestAPIEx2:



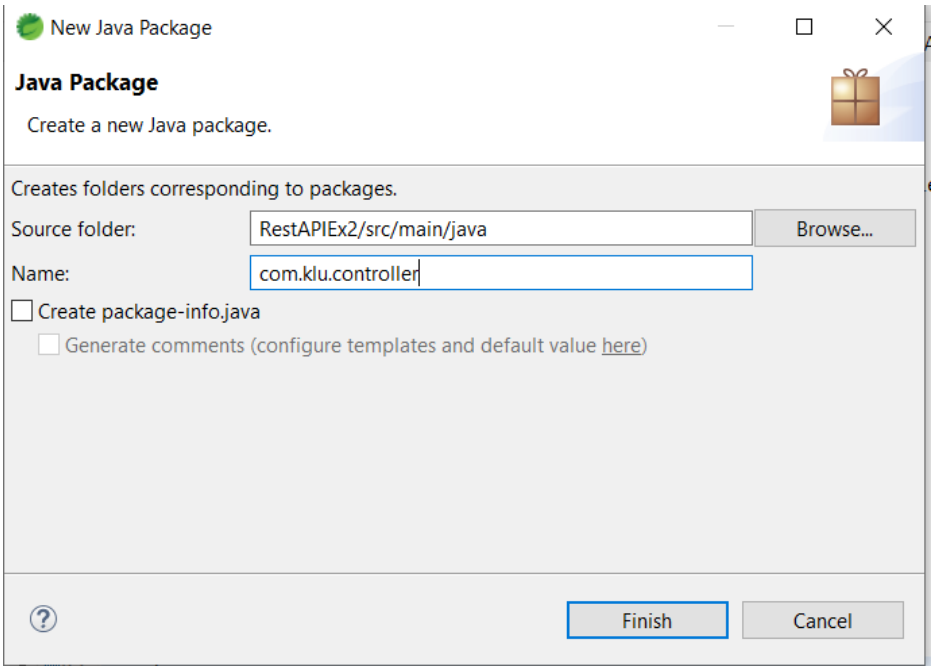

## Step-2:


New Spring Starter Project Dependencies
Spring Boot Version: 3.4.0
Frequently Used:
Eureka Discovery Client
Eureka Server
MySQL Driver
Spring Boot DevTools
Spring Data JPA
Spring Web
Available:
Type to search dependencies
AI
Developer Tools
Google Cloud
I/O
Messaging
Microsoft Azure
NoSQL
Observability
Ops
SQL
Security
Selected:
Spring Boot DevTools
Eureka Discovery Client
Spring Web
Make Default
Clear Selection
< Back
Next >
Finish
Cancel

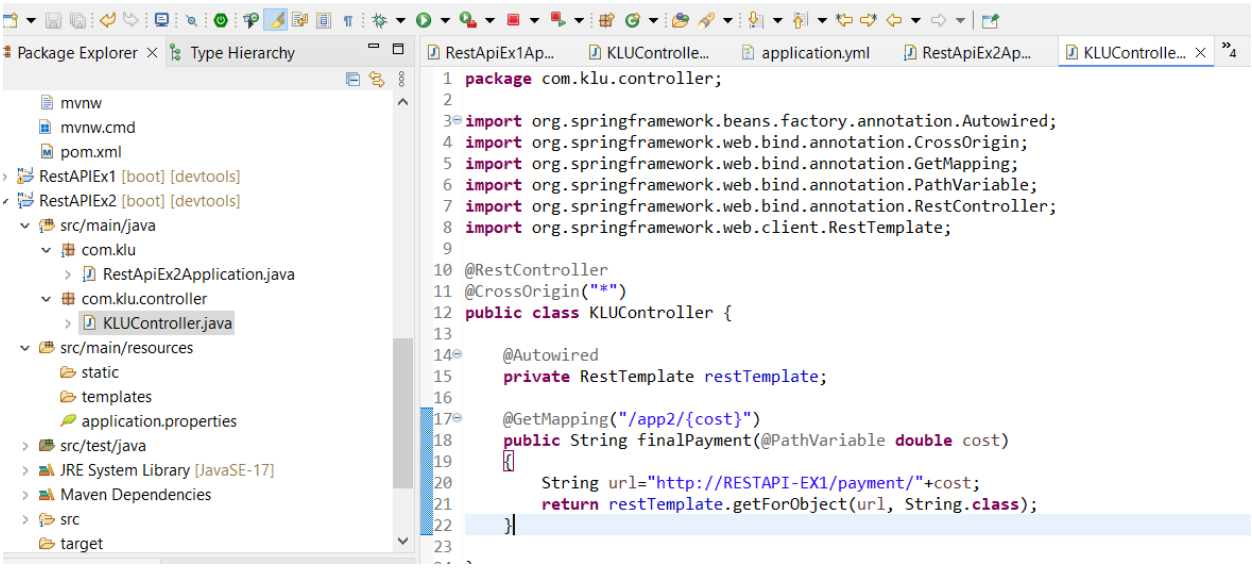
# Step-3:


File Edit Source Refactor Navigate Search Project Run Window Help
Package Explorer
Type Hierarchy
mvnw
mvnw.cmd
pom.xml
RestAPIEx1 [boot] [devtools]
RestAPIEx2 [boot] [devtools]
src/main/java
com.klu
RestApiEx2Application.java
src/main/resources
src/test/java
JRE System Library [JavaSE-17]
Maven Dependencies
src
target
HELP.md
mvnw
mvnw.cmd
pom.xml
Boot Dashboard
application.yml
RestApiEx1Ap...
KLUControlle...
application.yml
RestApiEx2Ap...
package com.klu;
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.cloud.client.discovery.EnableDiscoveryClient;
import org.springframework.cloud.client.loadbalancer.LoadBalanced;
import org.springframework.context.annotation.Bean;
import org.springframework.web.client.RestTemplate;
@SpringBootApplication
@EnableDiscoveryClient
public class RestApiEx2Application {
@Bean
@LoadBalanced
public RestTemplate getRestTemplate()
{
return new RestTemplate();
}
public static void main(String[] args) {
SpringApplication.run(RestApiEx2Application.class, args);
}
}

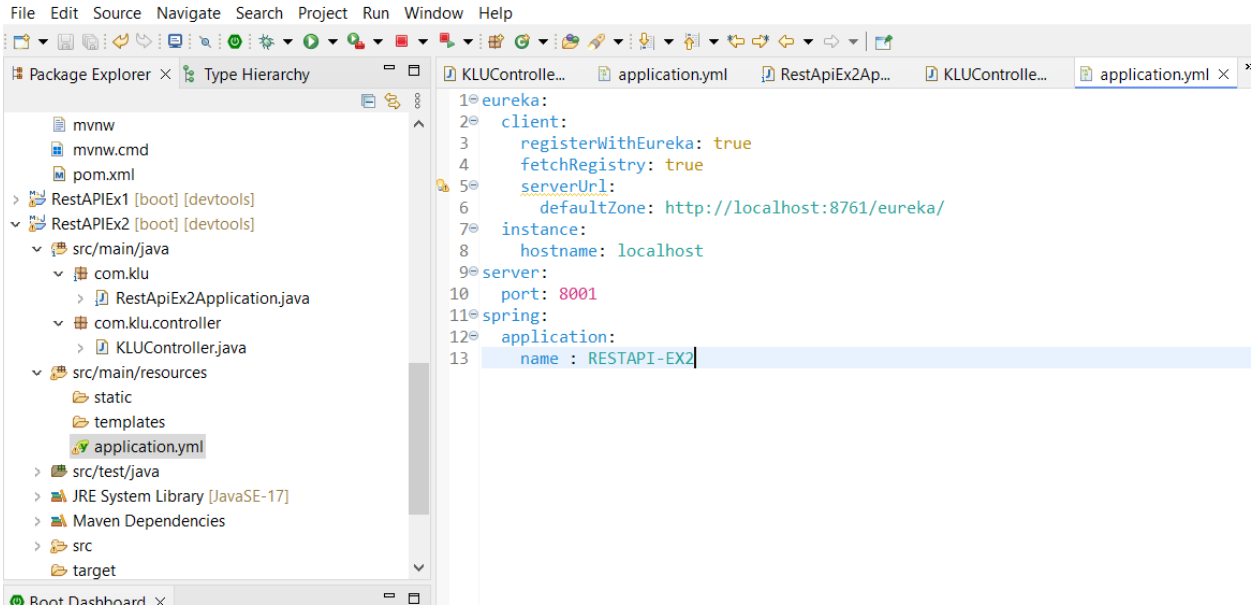
# Step-4


New Java Package
Java Package
Create a new Java package.
Creates folders corresponding to packages.
Source folder:
RestAPIEx2/src/main/java
Browse...
Name:
com.klu.controller
Create package-info.java
Generate comments (configure templates and default value here)
Finish
Cancel

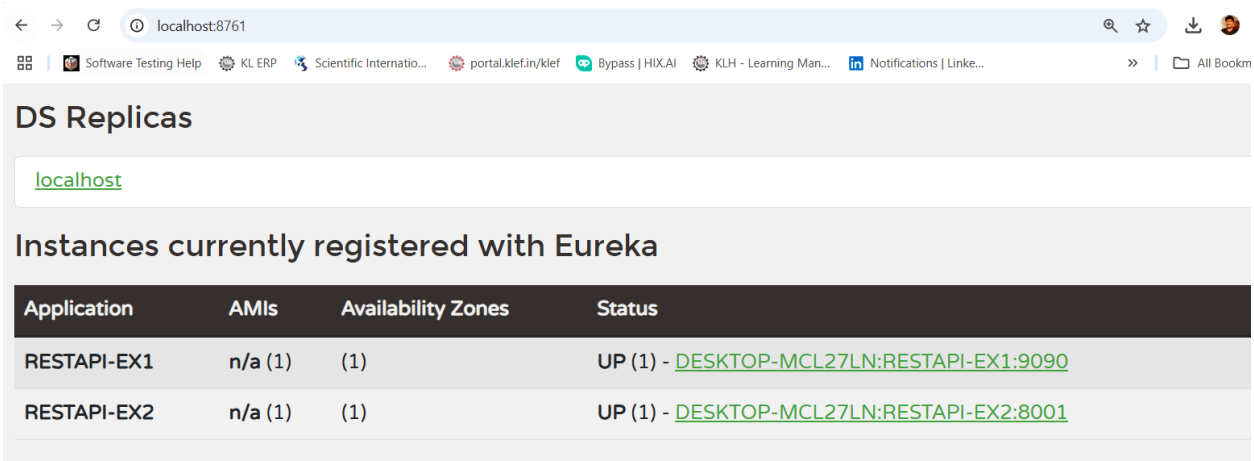
# Step-5


Package Explorer
Type Hierarchy
mvnw
mvnw.cmd
pom.xml
RestAPIEx1 [boot] [devtools]
RestAPIEx2 [boot] [devtools]
src/main/java
com.klu
RestApiEx2Application.java
com.klu.controller
KLUController.java
src/main/resources
static
templates
application.properties
src/test/java
JRE System Library [JavaSE-17]
Maven Dependencies
src
target
RestApiEx1Ap...
KLUControlle...
application.yml
RestApiEx2Ap...
KLUControlle...
package com.klu.controller;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.web.bind.annotation.CrossOrigin;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PathVariable;
import org.springframework.web.bind.annotation.RestController;
import org.springframework.web.client.RestTemplate;
@RestController
@CrossOrigin("*")
public class KLUController {
@Autowired
private RestTemplate restTemplate;
@GetMapping("/app2/{cost}")
public String finalPayment(@PathVariable double cost)
{
String url="http://RESTAPI-EX1/payment/"+cost;
return restTemplate.getForObject(url, String.class);
}

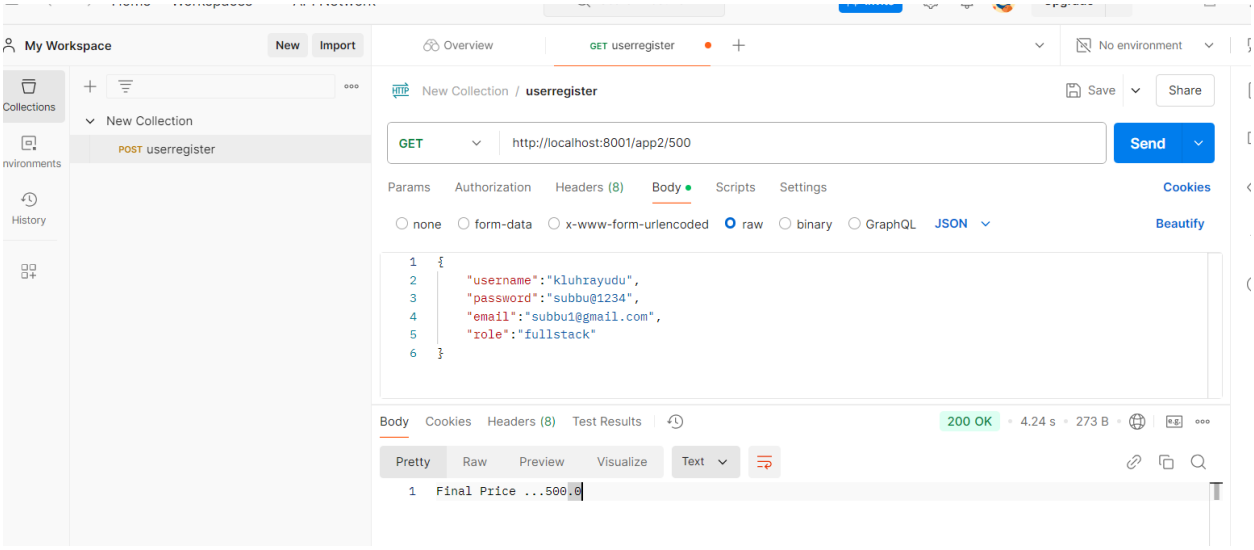
# Step-6


File Edit Source Navigate Search Project Run Window Help
Package Explorer
Type Hierarchy
mvnw
mvnw.cmd
pom.xml
RestAPIEx1 [boot] [devtools]
RestAPIEx2 [boot] [devtools]
src/main/java
com.klu
RestApiEx2Application.java
com.klu.controller
KLUController.java
src/main/resources
static
templates
application.yml
src/test/java
JRE System Library [JavaSE-17]
Maven Dependencies
src
target
Boot Dashboard
KLUControlle...
application.yml
RestApiEx2Ap...
KLUControlle...
application.yml
eureka:
client:
registerWithEureka: true
fetchRegistry: true
serverUrl:
defaultZone: http://localhost:8761/eureka/
instance:
hostname: localhost
server:
port: 8001
spring:
application:
name : RESTAPI-EX2

# Step-7


localhost:8761
Software Testing Help
KL ERP
Scientific Internatio...
portal.klef.in/klef
Bypass | HIX.AI
KLH - Learning Man...
Notifications | Linke...
All Bookm
DS Replicas
localhost
Instances currently registered with Eureka
Application
AMIs
Availability Zones
Status
RESTAPI-EX1
n/a (1)
(1)
UP (1) - DESKTOP-MCL27LN:RESTAPI-EX1:9090
RESTAPI-EX2
n/a (1)
(1)
UP (1) - DESKTOP-MCL27LN:RESTAPI-EX2:8001

# Step-8


My Workspace
New
Import
Overview
GET userregister
No environment
Collections
New Collection
POST userregister
History
New Collection / userregister
Save
Share
GET
http://localhost:8001/app2/500
Send
Params
Authorization
Headers (8)
Body
Scripts
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
JSON
Beautify
{
"username":"kluhrayudu",
"password":"subbu@1234",
"email":"subbu1@gmail.com",
"role":"fullstack"
}
Body
Cookies
Headers (8)
Test Results
200 OK
4.24 s
273 B
Pretty
Raw
Preview
Visualize
Text
Final Price ...500.0